Stochastic Gradient Descent for Smooth Convex Functions

Lecture: Oluwanifemi Adekanye. Scribe: Irene Berezin
CPSC 536C

1 Problem setup

Our goal is to minimize some continuous function $f(x)$. Recall one step of gradient descent:

$$x_{t+1} = x_t - \eta_t \nabla f(x_t),$$

where η_t is the step size. In machine learning, f is typically an average of losses over the data,

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x).$$

Computing the full gradient $\nabla f(x) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(x)$ requires processing all the data at each iteration, which can be computationally expensive: $O(nd)$ per iteration, and $O(Tnd)$ for T iterations. We can instead run stochastic gradient descent, which uses the gradient of a single loss function:

$$x_{t+1} = x_t - \eta_t g_t, \qquad \mathbb{E}_t[g_t] = \nabla f(x_t),$$

where $\mathbb{E}_t[\,\cdot\,]$ denotes expectation conditional on x_t (i.e. over the randomness at step t only). For example, if $f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x)$, sample i_t uniformly from $\{1, \ldots, n\}$ and take $g_t = \nabla f_{i_t}(x_t)$. Each iteration then costs $O(d)$, independent of n.

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and β-smooth, with minimizer x^*, and write $f^* = f(x^*)$. Let x_0 be fixed, and set

$$R = \|x_0 - x^*\|, \qquad D_t = \|x_t - x^*\|^2.$$

We will also assume f is a finite sum of n points. (But note, that this is only for convenience, as everything below holds for any stochastic oracle satisfying the two assumptions that follow) [2].

Assumption 1 (Unbiased gradients)**.** *Since i_t is drawn uniformly,*

$$\mathbb{E}\left[\nabla f_{i_t}(x)\right] = \sum_{i=1}^{n} \frac{1}{n} \nabla f_i(x) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(x) = \nabla f(x).$$

Assumption 2 (Bounded variance)**.** *There exists $\sigma^2 < \infty$ such that for all x,*

$$\mathbb{E}_i \|\nabla f_i(x) - \nabla f(x)\|^2 \leq \sigma^2.$$

2 The Stochastic Gradient Descent algorithm

Algorithm 1 Stochastic Gradient Descent (SGD)

Require: initial point $x_0 \in \mathbb{R}^d$, step sizes $\{\eta_t\}_{t=0}^{T-1}$, number of iterations T

1: **for** $t = 0, 1, \ldots, T-1$ **do**
2: Sample $i_t \sim \text{Uniform}\{1, \ldots, n\}$
3: $g_t \leftarrow \nabla f_{i_t}(x_t)$
4: $x_{t+1} \leftarrow x_t - \eta_t g_t$
5: **end for**
6: **return** $\bar{x}_T = \dfrac{\sum_{t=0}^{T-1} \eta_t x_t}{\sum_{t=0}^{T-1} \eta_t}$

Theorem 1. *Let f be convex and β-smooth, and suppose Assumptions 1–2 hold. If $\eta_t \le \frac{1}{2\beta}$ for all t, then*

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \;\le\; \frac{R^2}{\sum_t \eta_t} + \frac{\sigma^2 \sum_t \eta_t^2}{\sum_t \eta_t}.$$

3 Proof of SGD convergence

Tools

We use three facts. The first is the assumption written in terms of g_t:

$$\mathbb{E}_t \|g_t - \nabla f(x_t)\|^2 \le \sigma^2 \qquad \text{(error is bounded)} \tag{1}$$

$$\mathbb{E}_t[g_t] = \nabla f(x_t) \qquad \text{(unbiased)} \tag{2}$$

The other two are properties of f:

$$\|\nabla f(x)\|^2 \le 2\beta\big(f(x) - f^*\big) \qquad (\beta\text{-smoothness}) \tag{3}$$

$$\langle \nabla f(x),\, x - x^* \rangle \ge f(x) - f^* \qquad \text{(convexity)} \tag{4}$$

Step 1: bounding the second moment of g_t

$$\begin{aligned} \mathbb{E}_t \|g_t\|^2 &= \|\nabla f(x_t)\|^2 + \mathbb{E}_t \|g_t - \nabla f(x_t)\|^2 \\ &\le 2\beta\big(f(x_t) - f^*\big) + \sigma^2. \end{aligned} \tag{5}$$

To see the equality, write the stochastic gradient as the true gradient plus noise:

$$g_t = \nabla f(x_t) + \varepsilon, \qquad \varepsilon = g_t - \nabla f(x_t).$$

Expanding the square,

$$\|g_t\|^2 = \|\nabla f(x_t)\|^2 + 2\,\langle \nabla f(x_t),\, \varepsilon \rangle + \|\varepsilon\|^2 .$$

Taking $\mathbb{E}_t$, the cross term vanishes because $\nabla f(x_t)$ is fixed given x_t and $\mathbb{E}_t[\varepsilon] = 0$ by (2). So

$$\begin{aligned} \mathbb{E}_t \|g_t\|^2 &= \|\nabla f(x_t)\|^2 + \mathbb{E}_t \|\varepsilon\|^2 \\ &= \|\nabla f(x_t)\|^2 + \mathbb{E}_t \|g_t - \nabla f(x_t)\|^2 , \end{aligned}$$

and the inequality in (5) follows from (3) and (1).

Step 2: one-step progress

We want to show

$$\mathbb{E}_t[D_{t+1}] \le D_t - 2\eta_t(1-\beta\eta_t)\big(f(x_t)-f^*\big) + \eta_t^2\sigma^2. \tag{6}$$

Using $x_{t+1} = x_t - \eta_t g_t$,

$$\begin{aligned} D_{t+1} &= \|(x_t - x^*) - \eta_t g_t\|^2 \\ &= D_t - 2\eta_t \langle g_t,\, x_t - x^*\rangle + \eta_t^2 \|g_t\|^2 . \end{aligned}$$

Taking $\mathbb{E}_t$ (here x_t, D_t and η_t are fixed) and using (2),

$$\mathbb{E}_t[D_{t+1}] = D_t - 2\eta_t \langle \nabla f(x_t),\, x_t - x^*\rangle + \eta_t^2\, \mathbb{E}_t \|g_t\|^2 .$$

Applying (5) to the last term,

$$\mathbb{E}_t[D_{t+1}] \le D_t - 2\eta_t \langle \nabla f(x_t),\, x_t - x^*\rangle + 2\eta_t^2\beta\big(f(x_t)-f^*\big) + \eta_t^2\sigma^2.$$

Applying convexity (4) to the inner product (it enters with a minus sign, so a lower bound on it gives an upper bound here),

$$\begin{aligned} \mathbb{E}_t[D_{t+1}] &\le D_t - 2\eta_t\big(f(x_t)-f^*\big) + 2\eta_t^2\beta\big(f(x_t)-f^*\big) + \eta_t^2\sigma^2 \\ &= D_t - 2\eta_t(1-\beta\eta_t)\big(f(x_t)-f^*\big) + \eta_t^2\sigma^2, \end{aligned}$$

which is (6).

Step 3: choosing the step size

$$\eta_t \le \frac{1}{2\beta} \;\Rightarrow\; \eta_t\beta \le \frac{1}{2} \;\Rightarrow\; 1-\beta\eta_t \ge \frac{1}{2} \;\Rightarrow\; 2\eta_t(1-\beta\eta_t) \ge \eta_t.$$

Since $f(x_t) - f^* \ge 0$, we can replace $2\eta_t(1-\beta\eta_t)$ by the smaller η_t in (6):

$$\mathbb{E}_t[D_{t+1}] \le D_t - \eta_t\big(f(x_t)-f^*\big) + \eta_t^2\sigma^2.$$

Rearranging,

$$\eta_t\big(f(x_t)-f^*\big) \le D_t - \mathbb{E}_t[D_{t+1}] + \eta_t^2\sigma^2.$$

Taking full expectations (over $i_0, \dots, i_{t-1}$ as well; law of itterated expectations $\mathbb{E}\big[\mathbb{E}_t[\,\cdot\,]\big] = \mathbb{E}[\,\cdot\,]$),

$$\eta_t\, \mathbb{E}\big[f(x_t)-f^*\big] \le \mathbb{E}[D_t] - \mathbb{E}[D_{t+1}] + \eta_t^2\sigma^2.$$

Step 4: telescoping

Summing over $t = 0, \dots, T-1$, the D terms telescope:

$$\sum_{t=0}^{T-1} \eta_t\, \mathbb{E}\big[f(x_t)-f^*\big] \le D_0 - \mathbb{E}[D_T] + \sigma^2 \sum_{t=0}^{T-1} \eta_t^2.$$

Since $D_T \ge 0$, we have $-\mathbb{E}[D_T] \le 0$, and $D_0 = R^2$ is deterministic, so

$$\sum_t \eta_t\, \mathbb{E}\big[f(x_t)-f^*\big] \le R^2 + \sigma^2 \sum_t \eta_t^2.$$

Dividing both sides by $\sum_t \eta_t$,

$$\frac{\sum_t \eta_t\, \mathbb{E}\big[f(x_t)-f^*\big]}{\sum_t \eta_t} \le \frac{R^2}{\sum_t \eta_t} + \frac{\sigma^2 \sum_t \eta_t^2}{\sum_t \eta_t}. \tag{7}$$

Step 5: Apply Jensens

The left side of (7) is a weighted average of function values, but we want a bound on a single point. We can take the definition of convexity here extended to many points: with weights $\lambda_t = \eta_t / \sum_s \eta_s$ that are nonnegative and sum to one (Jensen's inequality),

$$f(\bar{x}_T) \leq \frac{\sum_t \eta_t f(x_t)}{\sum_t \eta_t}, \qquad \bar{x}_T = \frac{\sum_t \eta_t x_t}{\sum_t \eta_t}.$$

Subtracting f^* from both sides (and writing $f^* = \frac{\sum_t \eta_t f^*}{\sum_t \eta_t}$),

$$f(\bar{x}_T) - f^* \leq \frac{\sum_t \eta_t f(x_t)}{\sum_t \eta_t} - f^* = \frac{\sum_t \eta_t \big(f(x_t) - f^*\big)}{\sum_t \eta_t}.$$

Taking expectations,

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \leq \frac{\sum_t \eta_t \, \mathbb{E}\big[f(x_t) - f^*\big]}{\sum_t \eta_t},$$

■

4 Constant step size

Take $\eta_t = \eta \leq \frac{1}{2\beta}$. Then $\sum_t \eta_t = \eta T$ and $\sum_t \eta_t^2 = \eta^2 T$, so

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \leq \frac{R^2}{\eta T} + \eta \sigma^2.$$

The first term (bias) decays like $1/T$, as in GD. The second term does not decay, so with a constant step size, SGD will not converge to the minimum value but will instead *oscillate in a neighbourhood around the minimum*. [2]

The noise term $\eta\sigma^2$ does not depend on T, so with η fixed the bound can never reach 0. But what if η is not fixed? Is there an optimal value of η we can use? Notice that the right-hand side is a bias–variance tradeoff. The first term wants a big η, since big steps get us to x^* faster. The second term wants a small η, to keep the noise down.

Let us try $\eta = c/\sqrt{T}$:

$$\frac{R^2}{\eta T} + \eta \sigma^2 = \frac{R^2}{c\sqrt{T}} + \frac{c\,\sigma^2}{\sqrt{T}}.$$

Both terms now decay like $1/\sqrt{T}$. The value of c that makes the two terms equal is $c = R/\sigma$. But note that Theorem 1 requires $\eta \leq \frac{1}{2\beta}$, so we cap the step size and use

$$\eta = \min\left\{\frac{1}{2\beta}, \frac{R}{\sigma\sqrt{T}}\right\}.$$

This gives

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \leq \frac{2\beta R^2}{T} + \frac{2R\sigma}{\sqrt{T}}. \tag{8}$$

And, in fact, notice that if $\sigma = 0$ (as in the full batch size case), we recover exactly the $O(1/T)$ GD rate!

5 Comparison with non-smooth GD

The analysis above is almost identical to the analysis of subgradient descent for non-smooth convex functions, which uses exact subgradients. We primarily follow Bubeck's [1, Sect. 3.1] analysis of projected subgradient descent for Lipschitz functions[1]

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex, but not necessarily differentiable, and suppose each subgradient is bounded:

$$\|g\| \le L \qquad \forall\, g \in \partial f(x) \text{ and } x.$$

By the subgradient inequality and Cauchy-Schwartz, this implies that f is L-Lipschitz, since for any x, y and $g \in \partial f(x)$, $f(x) - f(y) \le \langle g,\, x - y\rangle \le L\,\|x - y\|$, and symmetrically with x and y swapped. Subgradient descent iterates

$$x_{t+1} = x_t - \eta_t g_t, \qquad g_t \in \partial f(x_t), \qquad t = 0, \dots, T-1,$$

and returns the same weighted average $\bar{x}_T = \frac{\sum_t \eta_t x_t}{\sum_t \eta_t}$ as Algorithm 1. We define $R = \|x_0 - x^*\|$ and $D_t = \|x_t - x^*\|^2$ the same way as before.

The definition of a subgradient, $f(y) \ge f(x_t) + \langle g_t,\, y - x_t\rangle$ for all y, evaluated at $y = x^*$, gives the non-smooth analogue of (4):

$$f(x_t) - f^* \le \langle g_t,\, x_t - x^*\rangle .$$

Substituting $g_t = (x_t - x_{t+1})/\eta_t$ and using

$$2\,\langle a,\, b\rangle = \|a\|^2 + \|b\|^2 - \|a - b\|^2$$

with $a = x_t - x_{t+1}$, $b = x_t - x^*$ (so $a - b = x^* - x_{t+1}$, and $\|a - b\|^2 = D_{t+1}$) gives us:

$$\begin{aligned} f(x_t) - f^* &\le \frac{1}{\eta_t}\,\langle x_t - x_{t+1},\, x_t - x^*\rangle \\ &= \frac{1}{2\eta_t}\Big(\,\|x_t - x_{t+1}\|^2 + D_t - D_{t+1}\Big) \\ &= \frac{1}{2\eta_t}\big(D_t - D_{t+1}\big) + \frac{\eta_t}{2}\,\|g_t\|^2, \end{aligned}$$

where the last line uses $x_t - x_{t+1} = \eta_t g_t$. Multiplying by $2\eta_t$ and using $\|g_t\| \le L$ yields

$$2\eta_t\big(f(x_t) - f^*\big) \le D_t - D_{t+1} + \eta_t^2 L^2. \tag{9}$$

Which is the deterministic counterpart to step 3. Comparing the two, L^2 plays the role of σ^2, and no step-size condition is needed, because there is no smoothness term to absorb (The whole cost of the $\eta_t^2||g_t||^2$ term sits in the additive error $\eta_t^2 L^2$)

Summing (9) over $t = 0, \dots, T-1$, the D terms telescope, and since $D_T \ge 0$ and $D_0 = R^2$, we get

$$2\sum_t \eta_t\big(f(x_t) - f^*\big) \le D_0 - D_T + L^2\sum_t \eta_t^2 \le R^2 + L^2\sum_t \eta_t^2.$$

Dividing by $2\sum_t \eta_t$ and applying Jensen's inequality exactly as in Step 5 gives us

$$f(\bar{x}_T) - f^* \le \frac{\sum_t \eta_t\big(f(x_t) - f^*\big)}{\sum_t \eta_t} \le \frac{R^2 + L^2\sum_t \eta_t^2}{2\sum_t \eta_t}.$$

[1]Bubeck specifically works with projected subgradient descent over a convex set $\mathcal{X}$, where $\mathcal{X}$ is contained in some euclidian ball : $y_{t+1} = x_t - \eta g_t$ with $g_t \in \partial f(x_t)$, followed by $x_{t+1} = \Pi_{\mathcal{X}}(y_{t+1})$ (You can think of this as calculating the gradient, then projecting it back onto the set $\mathcal{X}$ should the gradient point outside the set). However, we work on $\mathcal{X} = \mathbb{R}^d$, where the projection $\Pi_{\mathbb{R}^d}$ is the identity, so $x_{t+1} = y_{t+1}$ exactly.

Let's compare this form to what we found earlier with SGD. In the non-smooth case, the only bound is $\|g_t\| \leq L$. This also doesn't improve as we approach the optimum (take for example $f(x) = |x|$, where $g_t = 1 \; \forall x \neq 0$). Conversely, in the smooth stochastic case, $\mathbb{E}_t \|g_t\|^2$ can be decomposed into a noise term that doesn't shrink, σ^2, and a coefficient that does.[2]

With $\eta_t = \eta$, we have $\sum_t \eta_t = \eta T$ and $\sum_t \eta_t^2 = \eta^2 T$, so this is

$$f(\bar{x}_T) - f^* \leq \frac{R^2}{2\eta T} + \frac{\eta L^2}{2}.$$

The right side is minimized at $\eta = R/(L\sqrt{T})$, where both terms equal $RL/(2\sqrt{T})$, giving

$$f(\bar{x}_T) - f^* \leq \frac{RL}{\sqrt{T}}.$$

Notice that the constant-step bound has the same shape as Section 4. We have a $1/T$ term from R^2, and a non-decaying ηL^2 term in place of $\eta\sigma^2$. The key takeaway here is that **both non-smooth gradients and variance prevent $\eta_t g_t$ from vanishing.** As we saw in class, this is not the case with smooth GD, and the extra error term vanishes!

6 A few additional interesting SGD results

All of these theorems come from *Handbook of Convergence Theorems for (Stochastic) Gradient Methods*, by Garrigos and Gower.

We note that there are a few slight differences in the assumptions used here. In the smooth case, we assume each f_i is convex and β_i-smooth (not a generall β-smoothness condition on all of f) and writes $\beta_{\max} = \max_i \beta_i$. We also measure noise only at the optimum,

$$\sigma_f^* = \inf_{x^* \in \arg\min f} \mathbb{E}_i \|\nabla f_i(x^*)\|^2.$$

Since $\nabla f(x^*) = 0$, this is the variance of the stochastic gradient at x^*, so Assumption 2 at $x = x^*$ gives $\sigma_f^* \leq \sigma^2$.

6.1 Smooth and strongly convex

Recall that f is μ-strongly convex if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|^2$ for all x, y.

Theorem 2 ([3, Thm. 5.8]). *Let each f_i be convex and β_i-smooth, and let f be μ-strongly convex. Run SGD with a constant step size $0 < \eta < \frac{1}{2\beta_{\max}}$. Then for all $t \geq 0$,*

$$\mathbb{E}\|x_t - x^*\|^2 \leq (1 - \eta\mu)^t R^2 + \frac{2\eta\,\sigma_f^*}{\mu}.$$

A corollary of this is that with some clever choices of η and T, we can guarantee $\mathbb{E}(x_T - x^*) \leq \varepsilon$ for any target accuracy $\varepsilon > 0$. Specifically we can let:

$$\eta = \min\left\{\frac{\varepsilon\mu}{4\sigma_f^*}, \frac{1}{2\beta_{\max}}\right\} \quad \text{and} \quad T \geq \max\left\{\frac{4\sigma_f^*}{\varepsilon\mu^2}, \frac{2\beta_{\max}}{\mu}\right\} \log\left(\frac{2R^2}{\varepsilon}\right)$$

[2] Bubeck [1, Sect. 3.2]: under β-smoothness the gradients themselves go to zero near the optimum, since $\|\nabla f(x)\|^2 \leq 2\beta(f(x) - f^*)$ by (3), so a fixed step size is effectively "auto-tuned" and, one can afford to be much more aggressive. (Think back to the example of $f(x) = |x|$...)

6.2 Non-smooth and convex: stochastic subgradient descent

Theorem 3 ([3, Thm. 9.6])**.** *Under these assumptions above, and the same non-smooth GD assumptions from earlier, for any step sizes $\eta_t > 0$ and every $T \geq 1$,*

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \leq \frac{R^2 + L^2 \sum_t \eta_t^2}{2 \sum_t \eta_t}.$$

Theorem 4 ([3, Thm. 9.7, Cor. 9.8])**.** *With constant step size $\eta_t = \eta$ and $\bar{x}_T = \frac{1}{T}\sum_t x_t$,*

$$\mathbb{E}\big[f(\bar{x}_T)\big] - f^* \leq \frac{R^2}{2\eta T} + \frac{\eta L^2}{2},$$

and taking $\eta = \frac{R}{L\sqrt{T}}$ gives $\mathbb{E}[f(\bar{x}_T)] - f^* \leq \frac{RL}{\sqrt{T}}$, so $T \geq R^2L^2/\varepsilon^2$ iterations suffice for accuracy ε.

The interesting connection here is that non-smooth SGD exactly matches non-smooth deterministic GD.

References

[1] S. Bubeck. Convex Optimization: Algorithms and Complexity. *Foundations and Trends in Machine Learning*, 8(3–4):231–357, 2015.

[2] S. Vaswani. Lecture 8: Stochastic Gradient Descent. CMPT 409/981: Optimization for Machine Learning, Simon Fraser University, October 13, 2022. `https://vaswanis.github.io/409_981-F22/L8.pdf`

[3] G. Garrigos and R. M. Gower. Handbook of Convergence Theorems for (Stochastic) Gradient Methods. arXiv:2301.11235v3, 2024.